Calculus — Mock Board Paper

Mathematics · Class 12 Time: 1 hour Max. marks: 50

Q1. Differentiate $y = x^2 \cdot \sin x$ with respect to x.

Q2. Evaluate $\int x \cdot e^x \, dx$.

Q3. Find the maxima and minima of $f(x) = x^3 - 6x^2 + 9x + 1$.

Q4. Evaluate $\int_0^{\pi/2} \sin^2 x \, dx$.

Q5. Find the area bounded by $y = x^2$ and $y = x$.

Q6. Solve the differential equation $dy/dx = y/x$.